

3 “WE CAN TURN ANY IP CAMERA INTO A REAL-TIME, AI-POWERED MONITOR”

A Vellore-based startup claims to turn any camera into a system that detects tampering, tracks people, and sends real-time alerts across multiple locations.

Quantum Sharq, a Vellore-based private limited startup established in 2023, began with a clear goal: creating practical solutions that deliver measurable impact. The company initially focused on healthcare, where Dr M. Saravanan, CEO and CTO of Quantum Sharq and a PhD in computer vision, observed that physiotherapy clinics struggled to track patient progress accurately, relying on daily pain scores that were difficult to manage and analyse. Drawing on his expertise in computer vision and software, he developed a platform that converted routine patient data into actionable insights.

Building on this foundation, the company introduced Quantum Vision X, a device that adds intelligence to existing cameras through integrated hardware and software. The device connects directly to a network video recorder (NVR), with all cameras linked to a single NVR port. Once powered on, it transmits camera data to a receiver, such as a computer or laptop, delivering real-time insights through a web app, mobile app, WhatsApp notifications, and alerts. The system supports all camera types and NVRs, including analogue cameras and DVR setups, without restrictions; the only requirement is an internet connection.

Regarding cameras such as smart or Wi-Fi cameras with cloud or SD card storage, the device is not required. “It is designed primarily for multi-camera setups with existing NVR or DVR infrastructure, making it ideal for counting and monitoring in offices or large facilities. For larger deployments, multiple devices can be used while keeping the system cost-effective,”



explains Saravanan.

The system integrates AI and machine learning with in-house-assembled hardware, providing instant alerts through cloud or edge computing. “Using development boards such as Raspberry Pi, we apply encrypted AI code for real-time analytics without requiring entirely new camera installations. Unlike other solutions, our system works with any IP camera and can scale across multiple locations, offering flexibility for both private enterprises and government deployments,” says Saravanan.

Quantum Vision X can classify visitors by gender and age, automatically count heads, and even detect duplicate entries by the same individual. Quantum Facial leverages similar AI to provide facial attendance tracking, while Quantum HealthCare continues to streamline clinic workflows. AI models such as CNN and SSD, combined with OpenCV and TensorFlow pipelines, handle object detection, facial recognition, and mask detection efficiently.

On being asked about design challenges, Saravanan discloses, “Design challenges mainly involve networking and PCB design. The AI/ML software runs well on Raspberry Pi, but considerations such as single-layer

versus multi-layer PCBs for multiple components require support.”

When asked about testing and verification, the team explains that the devices were tested in their office lab over a period of six to nine months. The team conducted multiple scenarios, including inter-

rupting the camera view with red and yellow cloths, using hands to block it, cutting wires, disconnecting connections, and even disconnecting the internet. Regarding accuracy, the team claims detection remains reliable, although minor delays may occur in the communication protocol. For example, if a camera is interrupted at 11:42, the alert message is typically sent to the client within 15 to 45 seconds.

The startup has already delivered 47 units across its product variants, generating ₹1.9 million in revenue. All hardware is assembled at its Vellore facility, though certain components, such as SIM modules and development boards, are sourced from trusted vendors. Plans are underway to design proprietary PCB boards to reduce production costs by up to 50 per cent, further consolidating control over hardware quality.

When discussing current challenges, Saravanan said, “The current challenge is penetrating the market, that’s all. We tried social media for marketing, but we do not think this is purely marketing. It is more about convenience marketing; we have to reach out to customers, pitch the product, and demonstrate how it will support them.” **EFY**

—Nidhi Agarwal